transform (DFT) is defined as

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-jnk2\pi/N}, \quad k = 0, 1, 2, \ldots, N-1 \tag{6-8}$$

where X is the spectrum of x, and its synthesis equation or inverse DFT ($IDFT$) is

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]e^{jnk2\pi/N}, \quad n = 0, 1, 2, \ldots, N-1 \tag{6-9}$$

The DFT converts a finite (periodic), discrete signal in the time domain ($x[n]$) into a finite, discrete spectrum in the frequency domain ($X[k]$).

We can derive the DFT informally from the Fourier series of a periodic signal. Let's compare equation (6-8) and that for the coefficients of the Fourier series, equation (1-37), which I repeat here:

$$c_k = \langle f(t), e^{jk\omega_0 t}\rangle = \frac{1}{T}\int_0^T f(t)e^{-jk\omega_0 t}dt \tag{6-10}$$

First, notice that the DFT uses a summation within $[0, N-1]$ and the series uses an integral within one period T. This difference arises because here the signal is discrete and finite (and so we write $x[n]$ instead of $f(t)$). The factor of $1/T$ in (6-10) is equivalent to that of $1/N$ in (6-9), where that factor has been arbitrarily included in the IDFT instead of the DFT.

Now let's compare the basis (the complex exponential). The frequency here is also discretized. Remember that a sampled signal has a Nyquist frequency of $\omega_s/2$, or π radians/sample, where ω_s is the sampling rate, equivalent to 2π radians/sample. Let's discretize this signal's frequencies using N equally-spaced points. The step $\Delta\hat{\omega}$ between neighboring frequencies (in radians/sample) is

$$\Delta\hat{\omega} = \frac{2\pi}{N} \tag{6-11}$$

This is what equation (6-8) indicates within the sum: that whatever nk is, it will be a multiple of $2\pi/N$. The kth frequency in radians can be written as

$$k\Delta\hat{\omega} = k\frac{2\pi}{N}, \quad k = 0, 1, 2, \ldots, N-1 \tag{6-12}$$

Substituting $k\Delta\hat{\omega}$ in (6-12) for $k\omega_0$ in the Fourier series basis should convince you that the DFT is the discrete version of the Fourier series. The coefficient $X[k]$ is the magnitude of the spectrum of x at frequency $k2\pi/N$.

Table 6.1 illustrates how the step between frequencies changes as the signal duration and sampling rate (and thus number of samples) change. Increasing the signal duration, and therefore the number of samples, without changing the sampling rate (which keeps the Nyquist cutoff constant) decreases the step between frequencies (or, if you prefer to think of this another way, increases the frequency resolution). On the other hand, decreasing the sampling rate decreases the number of samples (assuming the signal duration stays constant) and the Nyquist cutoff, so the step between frequencies stays constant.